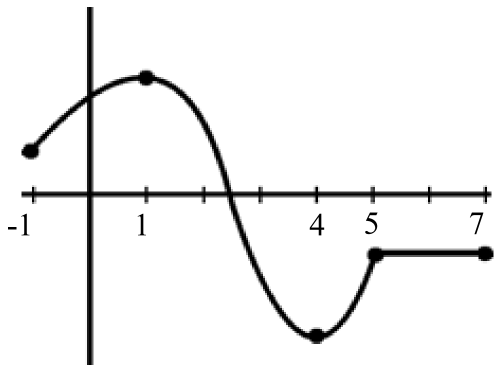
MTH 200 Chapter 4: Applications of Derivatives

4.5 Derivatives & the Shape of a Graph

Increasing/Decreasing:

General Idea:
Identify the intervals on which the function pictured below is increasing, decreasing, and constant.



Theorem: Let f be a function that is continuous $[a,b]$ and differentiable on (a,b).

(a) If $f'(x) > 0$ for every value of x in (a,b), then f is increasing on (a,b).

(b) If $f'(x) < 0$ for every value of x in (a,b), then f is decreasing on (a,b).

Note: A proof of this theorem requires the Mean Value Theorem. A proof is provided in the lecture notes.

<u>Example 1:</u> Determine on what intervals $f(x) = x^3 + 3x^2$ is increasing and decreasing.

<u>Question:</u> In the example above what would you expect is happening on the graph of the function when $x = -2$ and $x = 0$?

Theorem (The First Derivative Test): Suppose that *c* is a critical number of a continuous function *f*.

(a) If $f'(x)$ changes from positive to negative at *c*, then *f* has a local maximum at *c*.

(b) If $f'(x)$ changes from negative to positive at *c*, then *f* has a local minimum at *c*.

(c) If $f'(x)$ does not change (positive on both sides of *c* or negative on both sides of *c*) at *c*, then *f* has no local maximum or minimum at *c*.

<u>Example 2:</u> Determine on what intervals $f(x) = x^2 - 4x + 3$ is increasing and decreasing and state all local extrema of the graph.

Example 3: Suppose $f(x)$ is a function where $f'(x) = e^x (x+5)^2 (x^2 - 4)$. Find the intervals on which *f* is increasing and decreasing. Find the *x*-value(s) at where all local extrema occur.

Example 4: Determine on what intervals $f(x) = x^2 \ln(x)$ is increasing and decreasing and state all local extrema of the graph.

Concavity:

Definition: If f is differentiable on an open interval, then f is said to be

- ***concave up*** on the open interval if $f'(x)$ is increasing on the open interval.
- ***concave down*** on the open interval if $f'(x)$ is decreasing on the open interval.

This idea leads to the following theorem.

Theorem (Concavity Test):

(a) If $f''(x) > 0$ for all x in I, then the graph of f is concave upward on I.

(b) If $f''(x) < 0$ for all x in I, then the graph of f is concave downward on I.

Example 5: Find the intervals where Find the intervals where $f(x) = \frac{1}{4}x^4 - x^3$ is concave up/down.

Question: In Example 5, at what points did the concavity of the function change?

Definition: If f is a continuous function on an open interval containing the point P, and if f changes its concavity at the point at P, then we say that f has an **inflection point** at P.

<u>Example 6:</u> Determine the inflection points of the function $f(x) = \frac{1}{10}x^5 - \frac{1}{3}x^4 + \frac{1}{3}x^3 + 5$.

Theorem (The Second Derivative Test): Suppose $f''(x)$ is continuous near c.

(a) If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at c.

(b) If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at c.

<u>Example 7:</u> Use the Second Derivative Test to find all relative extrema of $f(x) = 3x^5 - 5x^3$.

Example 8: Find the intervals where $f(x)=x^3-3x+2$ is increasing, decreasing, concave up, and concave down. Find all local extrema and inflection points of *f*. Then, use this information to sketch a graph of the function.

Example 9:
Sketch a possible graph of a function f that satisfies the following conditions:

(i) $f'(x)>0$ on $(-\infty,1)$ and $f'(x)<0$ on $(1,\infty)$

(ii) $f''(x)>0$ on $(-\infty,-2)\cup(2,\infty)$ and $f''(x)<0$ on $(-2,2)$

(iii) $\lim_{x\to-\infty} f(x)=-2$ and $\lim_{x\to\infty} f(x)=0$